GRE Prep

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1 Multivariable Calculus

Here, we include no proofs.

Exercise 1.1. Find the equation of the plane in $\mathbb{R}^3$ passing through $P = (0, 0, 1), Q = (1, 0, 0), R = (1, 1, 1)$.

Solution: We use a reference point and use the vector cross product:

$$v := \overrightarrow{PQ} = \langle 1, 0, -1 \rangle, \qquad w := \overrightarrow{PR} = \langle 1, 1, 0 \rangle$$

Now, the normal vector to P is $v \times w$:

$$v \times w = \langle 1, -1, 1 \rangle$$

The equation of the plane is given by

$$\langle 1, -1, 1 \rangle \cdot \langle x - 0, y - 0, z - 1 \rangle,$$

which is simply $x - y + z = 1$. □

Exercise 1.2. Compute the arc length of the helix $\gamma(t) = (\sin(t), -\cos(t), t)$ from $t = 0$ to $t = 2\pi$.

Solution: We must use the arc-length formula $\int_t \|\gamma'(t)\| dt$.

$$\int_{t=0}^{2\pi} \sqrt{\cos^2(t) + \sin^2(t) + 1^2} dt = \int_{t=0}^{2\pi} \sqrt{2} dt = 2\pi\sqrt{2}.$$

□

The following is a nice way to determine if a limit doesn't exist:

Theorem 1.3. $\lim_{(x,y)\to(a,b)} f(x, y) = L$ *if and only if for all* $\gamma \colon [-, 1, 1] \to \mathbb{R}^2$ *such that* $\gamma(0) = (a, b)$, *then* $\lim_{t\to 0} f(\gamma(t)) = L$.

Exercise 1.4. Prove that $\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2}$ does not exist.

Solution: We consider the paths $\gamma(t) = (t, t)$ and $\delta(t) = (t, -t)$, which both tend to $(0, 0)$ as $t \to 0$. Using γ, we have as follows:

$$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2 + y^2} = \lim_{t\to 0} \frac{t^2}{t^2 + t^2} = \frac{1}{2}$$

On the other hand, δ gives:

$$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2 + y^2} = \lim_{t\to 0} \frac{-t^2}{t^2 + t^2} = -\frac{1}{2}$$

Therefore, the limit doesn't exist. □

Exercise 1.5. Find $\lim_{(x,y)\to(0,0)} \frac{xy^2}{x^2+y^2}$.

Solution: We use polar coordinates and change the limit to $r \to 0$.

$$\lim_{(x,y)\to(0,0)} \frac{xy^2}{x^2 + y^2} = \lim_{r\to 0} \frac{r^3 \cos\theta \sin^2\theta}{r^2} = \lim_{r\to 0} r \cos\theta \sin^2\theta$$

Now,

$$0 \leq |\lim_{r\to 0} r \cos\theta \sin^2\theta| \leq |\lim_{r\to 0} r| = 0,$$

so the limit is 0 via squeeze. □

1.1 Grad, Div, Curl

Recall that the gradient satisfies the following:

Proposition 1.6.

$$\nabla\phi = \left\langle \frac{\partial\phi}{\partial x}, \frac{\partial\phi}{\partial y}, \frac{\partial\phi}{\partial z} \right\rangle$$

Further, if we want the directional derivative of ϕ in the deriction of a unit vector $\hat{s}$, denoted $\frac{\partial\phi}{\partial s}$, we have

$$\frac{\partial\phi}{\partial s} = \hat{s} \cdot \nabla\phi.$$

Exercise 1.7. If $\phi = axy^2 + byz + cx^3z^2$, where a, b, c are constants, find $\nabla\phi$. Further, find the directional derivative of ϕ at the point $(1, 4, 2)$ in the direction towards the point $(2, 0 - 1)$.

Solution: First, we compute the gradient:

$$\nabla\phi = \left\langle ay^2 + 3cz^2z^2, 2axy + bz, by + 2cx^3z \right\rangle$$

Therefore, the gradient at $P = (1, 4, 2)$ is:

$$(\nabla\phi)_P = \langle 16a + 12c, 8a + 2b, 4b + 4c \rangle$$

Now, s is the difference between the two points, which is just $s = (2, 0, -1) - (1, 4, 2) = (1, -4, -3)$. Thus, $\hat{s}$ is

$$\hat{s} = \frac{\langle 1, -4, -3 \rangle}{\sqrt{26}}$$

Finally, the directional derivative $\frac{\partial\phi}{\partial s}$, is as follows:

$$\frac{\partial\phi}{\partial s} = (\nabla\phi \cdot \hat{s})_P = \frac{(16a + 12c) - 4(8a + 2b) - 3(4b + 4c)}{\sqrt{26}} = \frac{-16a - 20b}{\sqrt{26}}$$

□

Definition 1.8. Let F be a vector field.

- The *divergence* of F is the scalar product $\nabla \cdot F$, writing $\operatorname{div} F = \nabla \cdot F$.
- The *curl* of F is the vector product $\nabla \times F$, writing $\operatorname{curl} F = \nabla \times F$.

We recall some useful identities relating these different notions.

Theorem 1.9. *Let $\phi\colon \mathbb{R}^3 \to \mathbb{R}$ and F be a vector field (or more simply, a vector valued function).*

- $\operatorname{curl}(\nabla\phi) = 0$
- $\operatorname{div}(\operatorname{curl} F) = 0$
- $\operatorname{curl}(\operatorname{curl}(F)) = \nabla(\operatorname{div} F) - \nabla^2 F$

2 Linear Algebra

Here, we just include many problems that will be useful. Some exercises emphasize how some wishful statements do not hold.

Exercise 2.1. Show that $\{1, (x-1), (x-1)(x-2)\}$ is a basis of $\mathbb{P}_3[x]$ and that $W = \{p(x) \in \mathbb{P}_3[x] \colon p(1) = 0\}$ is a subspace of $\mathbb{P}_3[x]$. Find $\dim W$. Here, $\mathbb{P}_3[x]$ denotes the space of real polynomials of degree strictly less than 3.

Solution: The fact that this is a basis is clear. We find $\dim W$ using rank-nulity. The image has dimension 1, since two of the basis elements vanish. Thus, $\dim W = 2$. □

Exercise 2.2. Let $\alpha_1, \alpha_2, \alpha_3$, and α_4 be linearly independent. Answer true or false.

1. $\alpha_1 + \alpha_2, \alpha_2 + \alpha_3, \alpha_3 + \alpha_4, \alpha_4 + \alpha_1$ are linearly independent.
2. $\alpha_1 - \alpha_2, \alpha_2 - \alpha_3, \alpha_3 - \alpha_4, \alpha_4 - \alpha_1$ are linearly independent.
3. $\alpha_1 + \alpha_2, \alpha_2 + \alpha_3, \alpha_3 + \alpha_4, \alpha_4 - \alpha_1$ are linearly independent.
4. $\alpha_1 + \alpha_2, \alpha_2 + \alpha_3, \alpha_3 - \alpha_4, \alpha_4 - \alpha_1$ are linearly independent.

Solution: 1. False. We can make an easy relation among these elements.

2. False. Just sum them all.

3. True. Just expand.

4. False. Sum of last two is equal to difference of second and first. □

Exercise 2.3. Let $t \in \mathbb{R}$. Discuss the linear independence of the vectors over $\mathbb{R}$:

$$\alpha_1 = (1, 1, 0), \qquad \alpha_2 = (1, 3, -1), \qquad \alpha_3 = (5, 3, t).$$

Solution: We have the relations:

$$a + b + 5c = 0, \qquad a + 3b + 3c = 0, \qquad -b + tc = 0$$

The first two equations provide that $2b - 2c = 0$, or $b = c$. Thus, so long as $t \neq 1$, these are LI. More systematically, use the determinant condition. □

Exercise 2.4. Let W_1 and W_2 be nontrivial subspaces of a vector space V. The sum $W_1 + W_2$ is called a *direct sum*, denoted as $W_1 \oplus W_2$, if every element $\alpha \in W_1 + W_2$ can be uniquely written as

$$\alpha = w_1 + w_2,$$

where $w_1 \in W_1$ and $w_2 \in W_2$. Show that the following statements are equivalent:

1. $W_1 + W_2$ is a direct sum.
2. If $w_1 + w_2 = 0$ and $w_1 \in W_1$, $w_2 \in W_2$, then $w_1 = w_2 = 0$.
3. $W_1 \cap W_2 = \{0\}$.
4. $\dim(W_1 + W_2) = \dim W_1 + \dim W_2$.

How can the direct sum be extended to more than two subspaces?

Solution: Standard. □

Exercise 2.5. Let A, B, C be matrices. Prove if $r(A)$ denotes the rank of A, then prove the following:

1. $r(A+B) \leq r(A) + r(B)$.
2. $r(AB) \leq \min\{r(A), r(B)\}$.
3. $r(AB) + r(BC) - r(B) \leq r(ABC)$.

Solution:

□

Exercise 2.6. Let $A \in M_n(\mathbb{R})$.

1. Show that if $A^t = -A$ and n is odd, then $\det(A) = 0$.
2. Show that if $A^2 + I = 0$, then n is even. What is $A \in M_n(\mathbb{C})$ instead?

Solution: 1. We have $\det(A) = -\det(A)$, becaause n is odd, so $\det(A) = 0$.

2. We have $\det(A)^2 = (-1)^n$. Therefore, n must be even. If A is complex, then n can be even, because $\det(A) \in \mathbb{C}$ now. □

Exercise 2.7. Let $A, B \in M_n(\mathbb{R})$. Show that

$$\det \begin{pmatrix} A & B \\ -B & A \end{pmatrix} \geq 0.$$

State the analogue for complex matrices.

Solution: $\det(A)^2 + \det(B)^2$ is a sum of nonnegative reals, so is nonnegative. If we want a similar statement for complex matrices, then we just need to conjugate the bottom two:

$$\det \begin{pmatrix} A & B \\ \overline{-B} & \overline{A} \end{pmatrix} \geq 0$$

□

Exercise 2.8. Let $A, B \in M_n(\mathbb{C})$. Show that

$$\det \begin{pmatrix} A & B \\ B & A \end{pmatrix} = \det(A+B)\det(A-B)$$

Solution: Just use the fact that $\det(A^2 - B^2) = \det((A-B)(A+B)) = \det(A - B)\det(A+B)$. □

Exercise 2.9. For matrices of appropriate sizes, answer true or false:

1. If $A^2 = B^2$, then $A = B$ or $A = -B$.
2. If $r(A) = r(B)$, then $r(A^2) = r(B^2)$.

3. $r(A+kB) \leq r(A) + kr(B)$, where k is a positive scalar.

4. $r(A-B) \leq r(A) - r(B)$.

5. If $r(AB) = 0$, then $r(BA) = 0$.

6. If $r(AB) = 0$, then $r(A) = 0$ or $r(B) = 0$.

Solution: 1. False. Take $A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, B = I$.

2. Take $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, B = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$.

3. False, take $A - 0$, $B \neq 0$, and $k = \frac{1}{2}$. The correct statement instead
$$r(A+kB) \leq r(A) + kr(B).$$

4. Take $A = 0$, $B \neq 0$. Then, $r(-B)$ is positive, while $-r(B)$ is negative.

5. Take $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$. Then, $AB = 0$, while $BA = B \neq 0$.

6. Use the same example as 2. □

Proposition 2.10. *Show that $AB = I$ implies $BA = I$, where A, B are matrices over a field.*

Solution: This relies on some knowledge of algebra; particularly, about left or right Artinian rings. Note that the ring of matrices (over a field) form a finite dimensional algbera, and so they form an Artinian ring. We recall that left or right Artinian rings are rings where the DCC holds for left or right ideals. Let us use the left version. The claim we prove is that $ab = 1$ implies $ba = 1$, for any elements a, b (which satisfies the desired condition). We let the Artinian ring be called R. We note that $R \subseteq bR \subseteq b^2R \subseteq \dots$ is a descending chain of left ideals, and so $b^kR = b^{k+1}R$ for some positive integer k. In other words, $b^k = b^{k+1}c$, for some $c \in R$. Then, if we multiply by a^k on the left, we get that $bc = 1$. Then, $ba = babc = bc = 1$. □

Exercise 2.11. Let v_1, v_2, v_3, v_4 be vectors in $\mathbb{R}^2$, and consider the following six dot products:

$$v_1 \cdot v_2, \qquad v_1 \cdot v_3, \qquad v_1 \cdot v_4, \qquad v_2 \cdot v_3, \qquad v_2 \cdot v_4, \qquad v_3 \cdot v_4.$$

Which of the following statements about the six dot products could be true?

1. All six are negative.

2. Two are negative, and four are equal to 0.

3. Two are positive, and four are equal to 0.

Solution: Testing with some random vector might be good, but takes some clever guesses which I don't have the time to be searching for. The key is the dot product identity, which measures the angle depending on the sign. If positive, then acute, if 0, then perpendicular, if negative, then obtuse. Given four vectors in $\mathbb{R}^2$, then all six are negative if all vectors are obtuse with respect to each other. This is not possible. The remaining cases are possible. In the case 2, we situate them all perpendicularly to each other. In the case 3, we let $v_1 = v_2$, $v_3 = v_4$, and make these two vectors perpendicular to each other. □

3 Probability and Statistics

3.1 Probability via Area

Exercise 3.1. If x, y are randomly chosen in $[0, 1]$, what is the probability that $x \geq 2y$?

Solution: This probability can be determined by a ratio of areas. The total space has area 1. The area of the condition $x \geq 2y$ is as follows:

$$\int_{y=0}^{\frac{1}{2}} \int_{x=2y}^{1} dxdy = \int_{y=0}^{\frac{1}{2}} 1 - 2ydy = \frac{1}{2} - \left(\frac{1}{2}\right)^2 - \frac{1}{4}.$$

Thus, the probability is $\frac{1}{4}$. Note that in this case, it's better to just draw it out in $\mathbb{R}^2$ and use the triangle area formula. □

3.2 Basic Probability

Let Ω be the sample space and let $A, B \subseteq \Omega$ be events. Recall that the complement of A is $A^c = \Omega \setminus A$, and we have

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A).$$

We can also talk about unions and intersections of events. This gives us the principle of inclusion and exclusion (PIE):

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

Remark 3.2. If A and B are disjoint, then $\mathbb{P}(A \cap B) = 0$, so the above reads

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B).$$

3.3 Conditional Probability

Recall that we also had the notion of conditional probability. If $\mathbb{P}(B) > 0$, then

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Equivalently,

$$\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\mathbb{P}(B).$$

Symmetrically, if $\mathbb{P}(A) > 0$, then

$$\mathbb{P}(A \cap B) = \mathbb{P}(B \mid A)\mathbb{P}(A).$$

What is nice is that we can now define the notion of independence of events. We say that events A and B are independent if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

In the case that $\mathbb{P}(B) > 0$, conditional probability tells us that independence is equivalent to

$$\mathbb{P}(A \mid B) = \mathbb{P}(A).$$

Remark 3.3. Note that independence and disjointness are distinct. If A and B are disjoint and both have positive probability, then

$$\mathbb{P}(A \cap B) = 0,$$

while

$$\mathbb{P}(A)\mathbb{P}(B) > 0.$$

These lead to some very useful results:

Theorem 3.4 (Law of Total Probability)**.** *Suppose $B_1, \ldots, B_n$ form a partition of Ω. Then*

$$\mathbb{P}(A) = \sum_{i=1}^{n} \mathbb{P}(A \mid B_i)\mathbb{P}(B_i).$$

Theorem 3.5 (Bayes' Theorem)**.** *For events A and B with positive probability,*

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\mathbb{P}(A)}{\mathbb{P}(B)}.$$

Combining the two previous, we have Bayes's generalized theorem:

Theorem 3.6. *If $B_1, \ldots, B_n$ form a partition of Ω, then*

$$\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \mid B_j)\mathbb{P}(B_j)}{\sum_{i=1}^{n} \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)}.$$

3.4 Random Variables

Definition 3.7 (Discrete Random Variable)**.** A discrete random variable X assigns a real number to each outcome in the sample space. Its probability mass function (pmf) is

$$p_X(x) = \mathbb{P}(X = x),$$

where

$$\sum_{x} p_X(x) = 1.$$

Definition 3.8 (Continuous Random Variable)**.** A continuous random variable X has a probability density function (pdf) f_X satisfying

$$f_X(x) \geq 0$$

and

$$\int_{-\infty}^{\infty} f_X(x)dx = 1.$$

For $a < b$,

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x)dx.$$

Remark 3.9. For a continuous random variable,

$$\mathbb{P}(X = a) = 0.$$

We also have the notion of the cumulative distributive function.

Definition 3.10 (Cumulative Distribution Function)**.** The cumulative distribution function of X is

$$F_X(x) = \mathbb{P}(X \leq x).$$

Remark 3.11. For a continuous random variable, this means

$$F_X(x) = \int_{-\infty}^{x} f_X(t)dt,$$

where f_X is the pdf. Therefore, whenever F_X is differentiable,

$$f_X(x) = F_X'(x).$$

3.4.1 Expected Values of Random Variables

If X is a discrete random variable,

$$\boxed{\mathbb{E}[X] = \sum_x x\mathbb{P}(X = x)}.$$

If X is a continuous random variable,

$$\boxed{\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x)dx}.$$

Remark 3.12. More generally, for a function g,

$$\boxed{\mathbb{E}[g(X)] = \sum_x g(x)\mathbb{P}(X = x)}$$

in the discrete case, and

$$\boxed{\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x)\, dx}$$

in the continuous case.

Note that expectation is linear (for both discrete and continuous cases). That is, for any (real) constants a, b,

$$\boxed{\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]}.$$

Of course, more generally, this says

$$\mathbb{E}\left[\sum_{i=1}^{n} a_i X_i\right] = \sum_{i=1}^{n} a_i \mathbb{E}[X_i],$$

where the a_i are constants.

Remark 3.13. Note that independence is not required for the lienarity of expectation.

3.4.2 Variance

There is also the notion of variance. If we denote $\mu = \mathbb{E}[X]$, then

$$\boxed{\operatorname{Var}(X) = \mathbb{E}[(X-\mu)^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2}.$$

The second equality is justified by linearity. With this, we can define the standard deviation:

$$\boxed{\sqrt{\operatorname{Var}(X)}}.$$

Clearly, for constants a, b,

$$\boxed{\operatorname{Var}(aX+b) = a^2 \operatorname{Var}(X)}.$$

Particularly,

$$\operatorname{Var}(X+b) = \operatorname{Var}(X).$$

We can also define covariance of two random variables X and Y:

$$\boxed{\operatorname{Cov}(X,Y) = \mathbb{E}[(X-\mathbb{E}[X])(Y-\mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]}.$$

For arbitrary X and Y,

$$\boxed{\operatorname{Var}(X+Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\operatorname{Cov}(X,Y).}$$

If X and Y are independent, then we note that

$$\operatorname{Cov}(X,Y) = 0,$$

so

$$\boxed{\operatorname{Var}(X+Y) = \operatorname{Var}(X) + \operatorname{Var}(Y).}$$

Remark 3.14. Note that $\operatorname{Cov}(X,Y) = 0$ does not in general imply that X and Y are independent.

3.5 Common Distributions

Example 3.15 (Indicator Random Variable)**.** For an event A, define

$$I_A = \begin{cases} 1, & \text{if } A \text{ occurs,} \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$\mathbb{E}[I_A] = 1 \cdot \mathbb{P}(A) + 0 \cdot \mathbb{P}(A^c) = \mathbb{P}(A).$$

If

$$X = I_1 + \cdots + I_n$$

counts how many of a collection of events occur, then linaerity implies

$$\mathbb{E}[X] = \mathbb{E}[I_1] + \cdots + \mathbb{E}[I_n].$$

Example 3.16. If

$$X = \begin{cases} 1, & \text{with probability } p, \\ 0, & \text{with probability } 1-p, \end{cases}$$

then

$$\mathbb{E}[X] = p, \qquad \mathrm{Var}(X) = p(1-p).$$

Example 3.17 (Binomial Distribution)**.** If X counts the number of successes in n independent trials, each with success probability p, then $X \sim \mathrm{Bin}(n,p)$. Its pmf is

$$\mathbb{P}(X=k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

Moreover,

$$\mathbb{E}[X] = np, \qquad \mathrm{Var}(X) = np(1-p).$$

Example 3.18 (Geometric Distribution)**.** If X is the number of trials required to obtain the first success, then

$$\mathbb{P}(X=k) = (1-p)^{k-1}p, \qquad k = 1,2,\dots$$

and

$$\mathbb{E}[X] = \frac{1}{p}, \qquad \mathrm{Var}(X) = \frac{1-p}{p^2}.$$

The geometric distribution has the interesting property

$$\mathbb{P}(X > m+n \mid X > m) = \mathbb{P}(X > n).$$

Example 3.19 (Poisson Distribution)**.** If $X \sim \mathrm{Poisson}(\lambda)$, then

$$\mathbb{P}(X=k) = e^{-\lambda}\frac{\lambda^k}{k!}, \qquad k = 0,1,2,\dots$$

and

$$\mathbb{E}[X] = \lambda, \qquad \mathrm{Var}(X) = \lambda.$$

Example 3.20 (Uniform Distribution)**.** If X is uniformly distributed on $[a,b]$, then

$$f_X(x) = \frac{1}{b-a}, \qquad a \le x \le b.$$

Moreover,

$$\mathbb{E}[X] = \frac{a+b}{2}, \qquad \mathrm{Var}(X) = \frac{(b-a)^2}{12}.$$

4 ODEs

Here, we have some basic ways to solve ODEs. We first go through some basic linear ODEs, which I wish to be familiar with and make it just an easy computational exercise.

Example 4.1 (First-order linear ODE; Integrating Factors)**.** Consider

$$y' + p(x)y = q(x).$$

Define the integrating factor

$$I(x) = e^{\int p(x)dx}.$$

This gives

$$y(x) = e^{-\int p(x)dx}\left(\int e^{\int p(x)dx} q(x)dx + C\right) = \frac{1}{I(x)}\left(I(x)q(x)dx + C\right).$$

Note that this comes from multiplying the ODE by $I(x)$ and integrating the differentiated chain rule.

Exercise 4.2. Let $y(t)$ be the solution to the differential equation

$$y' + 2ty = e^{-t^2}\sin t$$

such that $y(0) = 0$. What is $y(\pi)$?

Solution: This is first order linear. The integrating factor is e^{t^2}. Thus,

$$y(t) = e^{-t^2}\left(\int e^{t^2}\cdot e^{-t^2}\sin t dt + C\right) = e^{-t^2}(-\cos(t) + C).$$

$y(0) = 0$, so $C = 1$. Thus, $y(\pi) = 2e^{-\pi^2}$. □

Example 4.3 (Second-order linear ODE with Constant Coefficients)**.** Consider the homogeneous equation

$$a_2y'' + a_1y' + a_0y = 0.$$

We consider $y = e^{\lambda x}$. Substitution gives the characteristic equation

$$a_2\lambda^2 + a_1\lambda + a_0 = 0.$$

If the characteristic roots are distinct $\lambda_1 \neq \lambda_2$, then

$$y(x) = c_1e^{\lambda_1 x} + c_2e^{\lambda_2 x}.$$

If the roots coincide, say to λ, then

$$y(x) = c_1e^{\lambda x} + c_2xe^{\lambda x}.$$

If the roots are complex, so $\lambda = \alpha \pm i\beta$, then the real-valued general solution is

$$y(x) = e^{\alpha x}\left(c_1\cos(\beta x) + c_2\sin(\beta x)\right).$$

Example 4.4 (Euler-Cauchy Equation)**.** A kth-order Euler-Cauchy equation has the form

$$\beta_k x^k\frac{d^ky}{dx^k} + \beta_{k-1}x^{k-1}\frac{d^{k-1}y}{dx^{k-1}} + \cdots + \beta_1 x\frac{dy}{dx} + \beta_0 y = f(x),$$

where $\beta_0, \ldots, \beta_k$ are constants. The key feature is that the coefficient of the jth derivative is proportional to x^j. For this, use $x = e^z$. This makes

$$x^j\frac{d^jy}{dx^j}$$

to be expressed as a linear combination of

$$\frac{d^j y}{dz^j}, \frac{d^{j-1} y}{dz^{j-1}}, \ldots, \frac{dy}{dz},$$

with constant coefficients, so the substitution $x = e^z$ transforms the Euler–Cauchy equation into a linear ODE with constant coefficients in the variable z. After solving the resulting constant-coefficient ODE, substitute $z = \ln x$ to write the solution in terms of x.

We see how this plays out below.

Exercise 4.5. Find a solution to the ODE

$$x^2 y'' + 3xy' + y = x^3.$$

Solution: We use $x = e^z$. We calculate the differentials first.

$$\frac{dy}{dx} = \frac{dy}{dz}\frac{dz}{dx} = \frac{1}{x}\frac{dy}{dz}$$

Next, we differentiate this to find $\frac{d^2y}{dx^2}$.

$$\frac{d^2 y}{dx^2} = \frac{d}{dx}\left(\frac{1}{x}\frac{dy}{dz}\right) = -\frac{1}{x^2}\frac{dy}{dz} + \frac{1}{x^2}\frac{d^2 y}{dz^2}.$$

Here, we used the chain rule:

$$\frac{d}{dx}\left(\frac{dy}{dz}\right) = \frac{d}{dz}\left(\frac{dy}{dz}\right)\frac{dz}{dx} = \frac{1}{x^2}\frac{d^2 y}{dz^2}$$

We now substitute back in, which gives the following ODE:

$$\frac{d^2 y}{dz^2} + 2\frac{dy}{dz} + y = e^{3z}.$$

This is in our realm now. We attain the following:

$$c_1 e^{-z} + c_2 z e^{-z} + \frac{1}{16} e^{3z}$$

We just substitute back $z = \ln(x)$ to get the desired form for $y(x)$. □

5 Complex Analysis

Here, we don't include proofs.

5.1 Basics of Holomorphic Functions

Theorem 5.1. *Let $f\colon \Omega \to \mathbb{C}$, where $\Omega \subseteq \mathbb{C}$ is open. The following statements are equivalent:*

- *f is holomorphic at $a \in \Omega$.*
- *f has a convergent power series $\sum_{n\geq 0} c_n (z-a)^n$ in an open disc centered at $a \in \Omega$.*

Definition 5.2. A complex-valued function is entire if it is holomorphic over $\mathbb{C}$.

Definition 5.3. Consider $f(z) = u + iv$ be a complex-valued function. Writing $z = x + iy$, the *Cauchy-Riemann equations* refer to the set of differential equations

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

Theorem 5.4. *Consider $f = u + iv \colon \Omega \to \mathbb{C}$, defomed on an open set $\Omega \subseteq \mathbb{C}$. If u and v are continuously differentiable and satisfy the Cauchy-Riemann equations on Ω, then f is holomorphic on Ω and $f'(z) = \frac{\partial f(z)}{\partial z}$.*

Example 5.5. $f(z) = \bar{z}$ is not holomorphic on $\mathbb{C}$. $\bar{z} = x - iy$, so $u'_x = 1 \neq -1 = v'_y$.

Exercise 5.6. Suppose that $f(x + iy) = u(x, y) + iv(x, y)$. If $u(x, y) = x^2 - y^2$ and $v(1, 1) = 2$. Find $v(4, 1)$.

Solution: The key is the FTC and the Cuachy-Riemann equations, because f is clearly holomorphic.

$$v(4,1) - v(1,1) = \int_1^4 v'_x(x,1)dx = \int_1^4 -u'_y(x,1)dy = \int_1^4 2dy = 6.$$

Therefore, $v(4, 1) = 8$. Note the use of the Cauchy Riemann equations. □

5.2 Cauchy's Integral Formula

Theorem 5.7 (Cauchy)**.** *Let f be holomorphic in an open set Ω, then f has infinitely many complex derivatives in Ω. Moreover, for simple, closed, piecewise-smooth curve $\gamma \subset \Omega$ and any z lying inside γ we have*

$$\frac{d^n f(z)}{dz^n} = \frac{n!}{2\pi i} \oint_\gamma \frac{f(\eta)}{(\eta - z)^{n+1}} d\eta.$$

Exercise 5.8. Evaluate the following integral, for $\gamma = \{z \colon |z| = 2\}$:

$$\frac{1}{2\pi i} \oint_\gamma \frac{e^z}{(z+i)(z-i)} dz$$

Solution: We have as follows:

$$\frac{1}{2\pi i} \oint_\gamma \frac{e^z}{(z+i)(z-i)} dz = \frac{1}{2\pi i} \frac{1}{2i} \oint_{\{z \colon |z|=2\}} \left(\frac{e^z}{z+i} - \frac{e^z}{z-i} \right) dz = \frac{1}{2i}(e^i - e^{-i}) = \sin 1.$$

□

Exercise 5.9. Let $C = \{z = 1 + e^{i\theta} \colon \theta \in [0, 2\pi]\}$. Calculate

$$\frac{1}{2\pi i} \int_C \left(\frac{\sin z}{z-1} \right)^2 dz.$$

Solution: Use Cauchy's integral formula with $n = 1$. We want the first derivative of $\sin^2(z)$, and plug in 1 into it. We have $2\sin(1)\cos(1) = \sin(2)$. □

Theorem 5.10 (Morera)**.** *Suppose f is a continuous function in the open disc D such that for any triangle T contained in D*

$$\int_T f(z)dz = 0,$$

then f is holomorphic.

Corollary 5.11 (Liouville)**.** *A bounded entire function is constant.*

5.3 Residue Theory

Definition 5.12. Let $f(z) = \sum_{-\infty}^{\infty} a_n(z-z_0)^n$, for $0 < |z-z_0| < R$ be the Laurent series for f at z_0. The *residue* of f at z_0 is $\text{Res}[f, z_0] = a_{-1}$.

Theorem 5.13. *Let f be holomorphic function inside and on a simple, closed, piecewise-smooth curve γ except at the singularities $z_1, \ldots, z_n$ in its interior. Then,*

$$\oint_\gamma f(z)dz = 2\pi i \sum_{j=1}^{n} \text{Res}[f, z_j]$$

Remark 5.14. Note that we can find the residue of f at a pole z_0 of order m as follows:

$$\text{Res}[f, z_0] = \lim_{z\to z_0} \frac{1}{(m-1)!}\frac{d^{m-1}}{dz^{m-1}}((z-z_0)^m f(z)).$$

Exercise 5.15. Evaluate $\oint_\gamma \frac{1}{z^5-z^3}dz$, where $\gamma = \{z\colon |z| = \frac{1}{2}\}$.

Solution: Let $f(z) = \frac{1}{z^5-z^3}$. The only relevant singularity in γ is $z = 0$. Thus, the integral is equal to $\text{Res}[f, 0]$, which is a pole of order 3:

$$\begin{aligned}
\lim_{z\to 0} \frac{1}{2!}\frac{d^2}{dz^2}\left(z^3 \cdot f(z)\right) &= \frac{1}{2}\lim_{z\to 0}\left(\frac{-2z}{(z^2-1)^2}\right)' \\
&= \frac{1}{2}\lim_{z\to 0}\frac{-2(z^2-1)^2 + 4z(z^2-1)\cdot 2z}{(z^2-1)^4} \\
&= -1
\end{aligned}$$

Thus, the integral evaluates to $-2\pi i$. □

6 Analysis and Point-Set Topology

We provide no proofs here.

6.1 Real Analysis

6.2 Topology

6.2.1 Open and Closed Sets

Lemma 6.1. *Let Y be a subspace of X. If U is open in Y and Y is open in X, then U is open in X.*

Remark 6.2. Note that if Y is not open in X, then there is no reason U should be open in X. As a counter example, consider $X = \mathbb{R}$ with the standard topology, $Y = [0, 1]$ (with the subpace topology), and $U = [0, \frac{1}{2})$. $U = (-\frac{1}{2}, \frac{1}{2}) \cap Y$, so is open in Y. However, clearly, U is not open in X, since opens in X are unions of open intervals.

Example 6.3. In the discrete topology on a set X, every subset of X is open. It follows that every set is closed as well.

Example 6.4. In any topology space X, $\emptyset$ and X are both closed and open.

A nice way to test if a set is closed in the subspace topology of some larger set is from the following:

Theorem 6.5. *Let $Y \subseteq X$. A is closed in Y if and only if it equals the intersection of a closed set of X with Y.*

It also follows then, from the lemma earlier, that:

Corollary 6.6. *Let $Y \subseteq X$. If A is closed in Y and Y is closed in X, then A is closed in X.*

There was also the notion of a closure, which is the intersection of all closed sets containing a set. Note that the resulting set is closed because arbitrary intersections of closed sets are closed.

Theorem 6.7. *Let $A \subseteq X$ be (only) a subset.*

- *$x \in \overline{A}$ if and only if every open set U containing x intersects A.*
- *If the topology of X is given by a basis, then $x \in \overline{A}$ if and only if every basis element containing x intersects A.*

Exercise 6.8. Let $X = \{n \in \mathbb{Z} \colon n \geq 2\}$ be the topoological space X with the topology generated by sets of the form $U_k := \{n \in X \colon n \mid k\}$, for $k \geq 2$, so that open sets of X are the empty set and arbitrary unions of the sets U_k. For any $n \in X$, determine the closure of the one point set $\{n\}$ in X.

Solution: Note that $m \in \overline{\{n\}}$ if and only if every open neighborhood of m meets $\{n\}$. In other words, we mean that if $m \in U_k$, then $\{n\} \cap U_k \neq \emptyset$, so $n \in U_k$. That is to say, if $m \mid k$, then we need $n \mid k$. Therefore, $m = ns$. Therefore, the closure is of the form $\{sn \colon s \in \mathbb{Z}_{>0}\}$. □

Definition 6.9. If A is a subset of the topological space X, then $x \in X$ is called a *limit point* of A if every neighborhood of x intersects A in some point other than x itself.

Remark 6.10. In other words, x is a limit point of A if x is in the closure of $A \setminus \{x\}$.

Theorem 6.11. *The closure of a set A in a topological space X is precisely the set A union with the set of limit points of A in X.*

6.2.2 Hausdorff

Recall the notion of Hausdorff:

Definition 6.12. A topological space X is *Hausdorff* if for each pair of distinct points $x, y \in X$, there are neighborhoods of x, y, respectively, which are disjoint.

Theorem 6.13. *Every finite point set in a Hausdorff space is closed.*

Theorem 6.14. *If X is a Hausdorff space, then a sequence of points of X converges to at most one point of X.*

Proposition 6.15. *Products and subspaces of Hausdorff spaces are Hausdorff.*

6.3 Miscellanious

7 Group Theory

7.1 Group Actions and Orbit-Stabilizers

Let G be a group acting on a set X.

Definition 7.1. The *stabilizer* of $x \in X$ of a group action G, G_x, is the set $\{g \in G \colon gx = x\}$. We denote the *orbit* of which contains x as $\mathcal{O}_x$.

Proposition 7.2. *G_x is a subgroup of G.*

Proof. Clear. □

Proposition 7.3. *Let X be finite. Then, $|\mathcal{O}_x| = [G : G_x]$.*

Proof. We define a bijective function between the cosets G/G_x and $\mathcal{O}_x$. We define $\phi \colon \mathcal{O}_x \to G/G_x$ as $y \mapsto gG_x$, where $gx = y$. Note that such a $g \in G$ exists because $y \in \mathcal{O}_x$. This is also well-defined in the sense that $\phi(y)$ is not dependent on g (easy check). The claim is that this is bijective, but this is, again, a direct check. □

Proposition 7.4 (The Class Equation)**.** *Let X be finite. Let $X_G = \{x \in X \colon gx = x, \forall g \in G\}$ be the fixed points of X. Then,*

$$|X| = |X_G| + \sum |\mathcal{O}_{x_i}|,$$

where the x_i are the representatives of each orbit.

Proof. This is clear. Elements not fixed by G are in some orbit, while those fixed by G are in X_G. □

Corollary 7.5. *Let G be a finite group and act on itself via conjugation. Then,*

$$|G| = |Z(G)| + \sum n_i,$$

where each n_i denotes the cardinality of each nontrivial conjugacy class. Further, n_i divides $|G|$ for all i.

Proof. The summation follows from the theorem above. The size of each orbit is also equal to the size of the cosets, which divide $|G|$ by Lagrange. □

Theorem 7.6. *Let $|G| = p^n$, for p prime. Then, G has nontrivial center.*

Proof. We let G act on itself via conjugation. Then, the previous proposition says

$$|G| = |Z(G)| + \sum n_i,$$

where each n_i is the size of the conjugacy classes. Then, $p \mid n_i$ and solution $p \mid |Z(G)|$. The identity is always in the center, so $|Z(G)| \geq 1$. Therefore, $|Z(G)| \geq p$ and so there is some nontrivial element in the center. □

Corollary 7.7. *Every group of order p^2 is abelian.*

Proof. By the previous theorem, if the center has order p^2, then clearly the group is abelian. Suppose not then. It follows that $Z(G)$ and $G/Z(G)$ both have order p, which are cyclic. Let $aZ(G)$ generate $G/Z(G)$. Then, we can write $gZ(G), hZ(G)$ as $g = a^m x, h = a^m y$ for some $x, y \in Z(G)$ and m, n. We just check for commuting of h and g, but this follow directly. □

Notation 7.8. We write $x \sim y$ if x and y belond to the same orbit.

Proposition 7.9. *If $x \sim y$, for $x, y \in X$, then G_x and G_y are iomorphic (as groups).*

Proof. Let $y = gx$, for some $g \in G$. Then, the map $\phi \colon G_x \to G_y$ defined as $a \mapsto gag^{-1}$ defines a bijective group homomorphism (thus, a group isomorphim). □

Theorem 7.10 (Burnide's Counting Lemma)**.** *Let k denote the number of orbit of X. We have as follows:*

$$k = \frac{1}{|G|} \cdot \sum_{g \in G} |X_g|.$$

Proof. The cardinality of elements of G which fix x is the sum $\sum_{g \in G} |X_g|$. Dually, viewed in terms of the stabilizer subgroups, this number is equal to the sum $\sum_{x \in X} |G_x|$. We also note that

$$\sum_{y \in \mathcal{O}_x} |G_y| = |\mathcal{O}_x| \cdot |G_x| = |G|.$$

Note that the first part follows by the previous lemma and the second equality follows by Lagrange. Now, we sum over the k distinct orbits to deduce that

$$\sum_{g \in G} |X_g| = \sum_{x \in X} X = k \cdot |G|.$$

□

7.2 Sylow's Theorems

Definition 7.11. A group G is a *p-group* if every element in G has as its order a power of p, for a prime p. A subgroup of a group is a *p-subgroup* if it is a p-group.

Recall Cauchy's theorem.

Theorem 7.12 (Cauchy)**.** *Let G be a finite group be a finite group with p dividing the order $|G|$. Then, G contains a subgroup of order p.*

Proof Sketch. Induct on the order of the group. The base case is immediate. Assume now that the result holds for all groups of order less than n (and p must divide n by hypothesis). We use the class equation and case work.

If the order of some conjugacy class is divisible by p, then we are done by the inductive hypotehsis.

Now, suppose that the order of no conjugacy class has order divisible by p. In this case, $[G : C_i]$ are all diviible by p (where C_i are the conjugacy classes), and the class equation implies that p divides the center. Now, $Z(G)$ is a finite abelian group, so the structure theorem implies that there is an element of order p in $Z(G)$. □

Corollary 7.13. *If G is finite, then G is a p-group if and only if $|G| = p^n$.*

The point of the Sylow theorem is to provide a partial converse to Cauchy's theorem.

Theorem 7.14 (Sylow I)**.** *Let G be finite and p be a prime such that p^r divides $|G|$. Then, G contains a subgroup of order p^r.*

Proof. Theorem 15.3 in [1]. □

Definition 7.15. A Sylow p-subgroup of a group G is a maximal p-subgroup of G.

Theorem 7.16 (Sylow II)**.** *Let G be a finite group and p be a prime dividing $|G|$. Then all Sylow p-subgroups of G are conjugate to each other.*

Proof. Theorem 15.6 in [1]. □

Theorem 7.17 (Sylow III)**.** *Let G be a finite group and let p be a prime dividing the order of G. Then the number of Sylow p-subgroups is congruent to* $1 \pmod{p}$ *and divides $|G|$.*

Definition 7.18. A *simple group* is one with no proper nontrivial normal subgroups.

Bibliography

[1] Thomas W. Judson and Stephen F. Judson. *Abstract Algebra: Theory and Applications*. Open-source textbook available at http://abstract.ups.edu/. Orthogonal Publishing L3C, 2020.